# Chapter 1: Energy & Respiration

## Key Outcomes:

- Why Living organisms need energy
- ATP Structure and Why its the universal currency for energy, How ATP is made
- Structure of mitochondria
- The 4 Stages of Aerobic Respiration
- Role of Coenzymes in Aerobic Respiration
- Difference in relative energy values of different respiratory substrates (lipids, carbohydrates, proteins) RQ Value (and calculations)
- Anaerobic Respiration pathways and the differences between them
- Adaptations of rice plants to be able to survive in anaerobic conditions

## Definitions

**Respiration: The process by which cells release energy from organic molecules, measured as ATP production from ADP and Pi.**

**ATP (Adenosine Triphosphate): The universal energy currency of cells, consisting of adenine, ribose sugar, and three phosphate groups. Energy is stored in the phosphate bonds.**

**Anabolic Reactions: reactions that result in building new molecules/compounds (e.g. Protein synthesis, DNA replication, etc...)**

**Aerobic Respiration: Respiration using oxygen as the final electron acceptor. More efficient than anaerobic respiration, producing ≈32 ATP per glucose.**

**Anaerobic Respiration: Respiration without oxygen. Produces 2 ATP per glucose and allows glycolysis to continue when oxygen is scarce.**

**Respiratory Quotient (RQ): The ratio of $CO_2$ produced to $O_2$ consumed during respiration (RQ = $CO_2$ out / $O_2$ in). Varies by substrate**

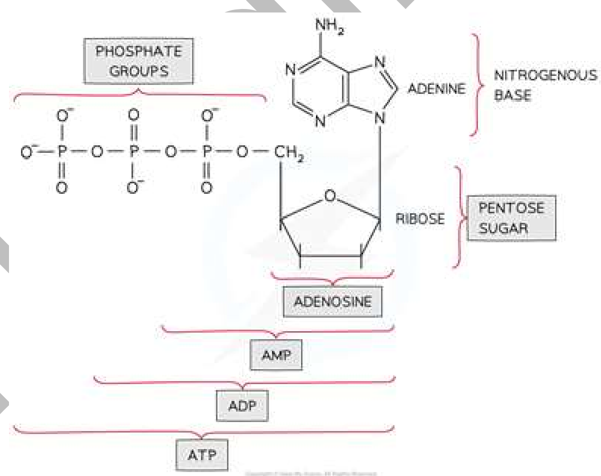
# Introduction

## Why do living organisms need energy?

**For vital processes within the body essential for life, like:**

1. Anabolic Reactions e.g. Protein synthesis, DNA replication (see definition above)
2. Active Transport
3. Muscle Contractions for movement
4. Exocytosis/Endocytosis
5. Transmission of nerve impulses

## What is ATP?




Adenosine Triphosphate, Its a molecule that is used as **a universal energy currency**

**ATP structure:**

- Adenine nitrogen base
- Ribose sugar
- 3 Phosphates (phosphorus + 4 oxygen)

Q: How does the structure of ATP differ from Adenine?

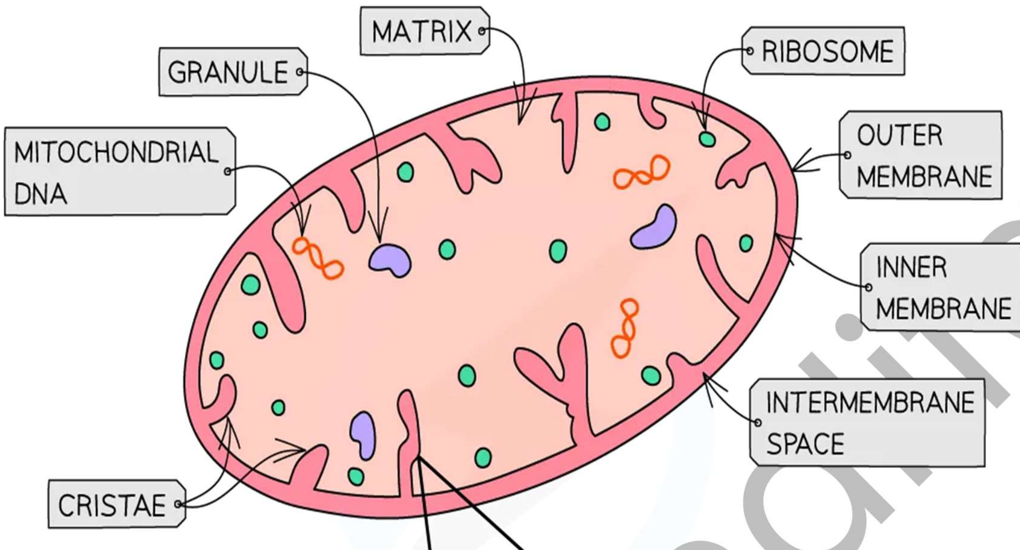
- **ATP has ribose not deoxyribose**
- **ATP has 3 phosphates**

Q: Describe the features that make ATP a suitable universal currency for energy

VERY IMPORTANT QUESTION!!

- Small water soluble molecule
- Hydrolysis leads to **immediate energy release**
- Reversible reaction (ATP $\rightleftharpoons$ ADP + Pi + 30.5kJ)
- Has **High Turnover** (hydrolysis of each phosphate releases **30.5kJ**)
- Link between energy yielding and energy requiring reactions

# Mitochondria Structure




## Outer membrane:

Permeable to several small molecules (as it contains many transport proteins)

## Inner membrane:

- Less permeable than outer membrane (naturally not permeable to protons)
- Site of ETC and ATP synthase (used in oxidative phosphorylation)
- Folded forming Cristae

## Intermembrane space:

- Space between the two membranes
- Accumulates Protons ($H^+$). So, its more acidic than the Matrix

## Mitochondrial matrix:

- Like a cytoplasm, but for the mitochondria, contains things like:
  - **Mitochondrial DNA:** Carries genes of necessary proteins and enzymes within a mitochondrion
  - **70S Ribosomes:** For Protein synthesis of respiratory enzymes and proteins within a mitochondrion

*Q: Explain the importance of the inner membrane being folded into cristae

- **Increases the surface area** of inner membrane
- SO **Increased number of ETC and ATP synthase**
- So **increases ATP production**

## A* Student Question:

Q: Explain why when placed in water, an animal cell bursts while a mitochondrion doesn't burst

- SINCE The mitochondrion **has foldings** in the inner membrane **(cristae)**
- Allowing it to **expand** when water enters/when it becomes turgid
- Animal cell doesn't have a cell wall to protect it from bursting

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# Aerobic Respiration

### How is ATP made?

**ATP is made by phosphorylating ADP (ADP + Pi → ATP)**

**SO, this means that cells have a constant supply of ADP OR ATP at all times**

**Throughout the stages of respiration, ATP is made from different reactions and in different amounts.**

1. Substrate linked/level phosphorylation:

(reactions where ATP is a Biproduct of the reaction), This happens in these stages of respiration

   a. **Glycolysis**
   b. **Krebs Cycle**

2. **Oxidative Phosphorylation**

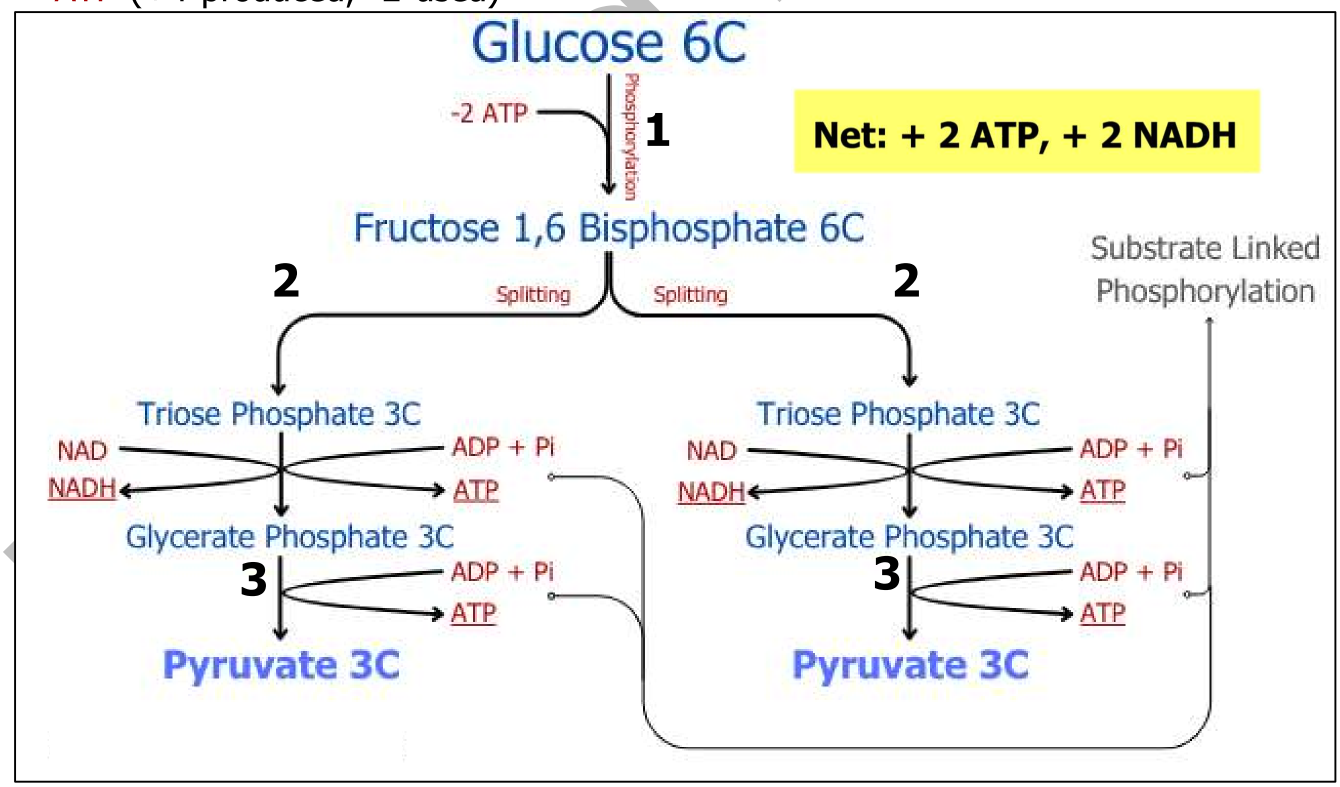
a stage in Aerobic respiration that results in producing large amounts of ATP utilizing **ETC, chemiosmosis and ATP Synthase** protein (will be explained further in detail)

Note: Different Substances (carbohydrates, lipids, protein) can be **substrates for respiration** (*REMEMBER THIS FOR LATER*)

# Stages of Aerobic Respiration (Carbohydrates)

## 1. Glycolysis (Cytoplasm)

1. **Phosphorylation**: Glucose (6C) is phosphorylated using 2 ATP to form fructose-1,6-bisphosphate (6C), this is to:
   a. **Trap the Glucose molecule in the cell (prevent it from diffusing out)**
   b. **Activate the Glucose/Increase its energy levels**
2. **Splitting:** Fructose-1,6-bisphosphate splits into TWO 3C Triose phosphates (TP).
3. **Oxidation:** Each Triose phosphate is oxidized to form Glycerate Phosphate (GP), reducing NAD to NADH and producing ATP. Then GP is broken down into Pyruvate (3C) producing ATP.
4. **ATP Production:** Substrate-linked phosphorylation yields a net gain of +2 ATP (+4 produced, -2 used)




***Pyruvate molecules then diffuse into the mitochondria where they continue aerobic respiration stages***

Q: Explain why glycolysis needs ATP first/ why Glucose gets phosphorylated in the beginning of glycolysis

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## Slow down! What is this NAD and NADH??

**NAD is a <u>coenzyme</u> that is ESSENTIAL for respiration, since it:**

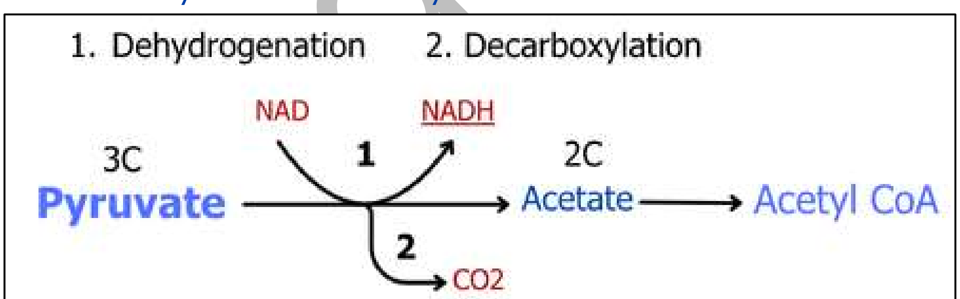
- Carries H atoms from dehydrogenation reactions, gets reduced into reduced NAD (NADH)
- Acts as a **Coenzyme to Dehydrogenase** enzyme, responsible for dehydrogenation
- **Transports Hydrogen** Atoms **to Cristae** of mitochondria for ETC (Oxidative phosphorylation)
- Each NADH results in production of 2.5 ATP molecules in Oxidative Phosphorylation

***FAD Is another Coenzyme just like NAD, but only 1.5 ATP is produced from it in Oxidative phosphorylation***

## 2. Link Reaction (Mitochondrial Matrix)

1. Decarboxylation: Pyruvate loses a $CO_2$ molecule to form an acetyl group, becoming a 2C molecule (by decarboxylase enzyme)
2. Dehydrogenation: Pyruvate is oxidized by dehydrogenase enzyme (pyruvate - H), reducing NAD to NADH, So Pyruvate → Acetate
3. <u>Coenzyme A</u> Binding: The acetyl group combines with coenzyme A to form Acetyl-CoA, to <u>join Link Reaction with Kreb's Cycle</u>

*Link Reaction: 2 Pyruvate → 2 Acetyl CoA + 2 NADH




**Total So Far:**

**Glucose → 2 Acetyl CoA + 4 NADH + 2 ATP**

**Exam Tip: The link reaction indirectly requires oxygen because NADH must be reoxidized in the ETC to regenerate NAD.**

**So, No oxygen = No link reaction**

**(+no Krebs cycle, oxidative phosphorylation)**

**Chemistry Reminder:**

- Reduction = Gain of hydrogen atom
  NAD + H —> NADH
- Oxidation = loss of hydrogen atom (Regeneration of NAD)
  NADH —> NAD + H

## 3. Krebs Cycle (Mitochondrial Matrix)

**Remember**, 2 Acetyl CoA are produced from link reaction, each enter separate Krebs cycle, so 2x Krebs Cycles occur in respiration of 1 Glucose molecule

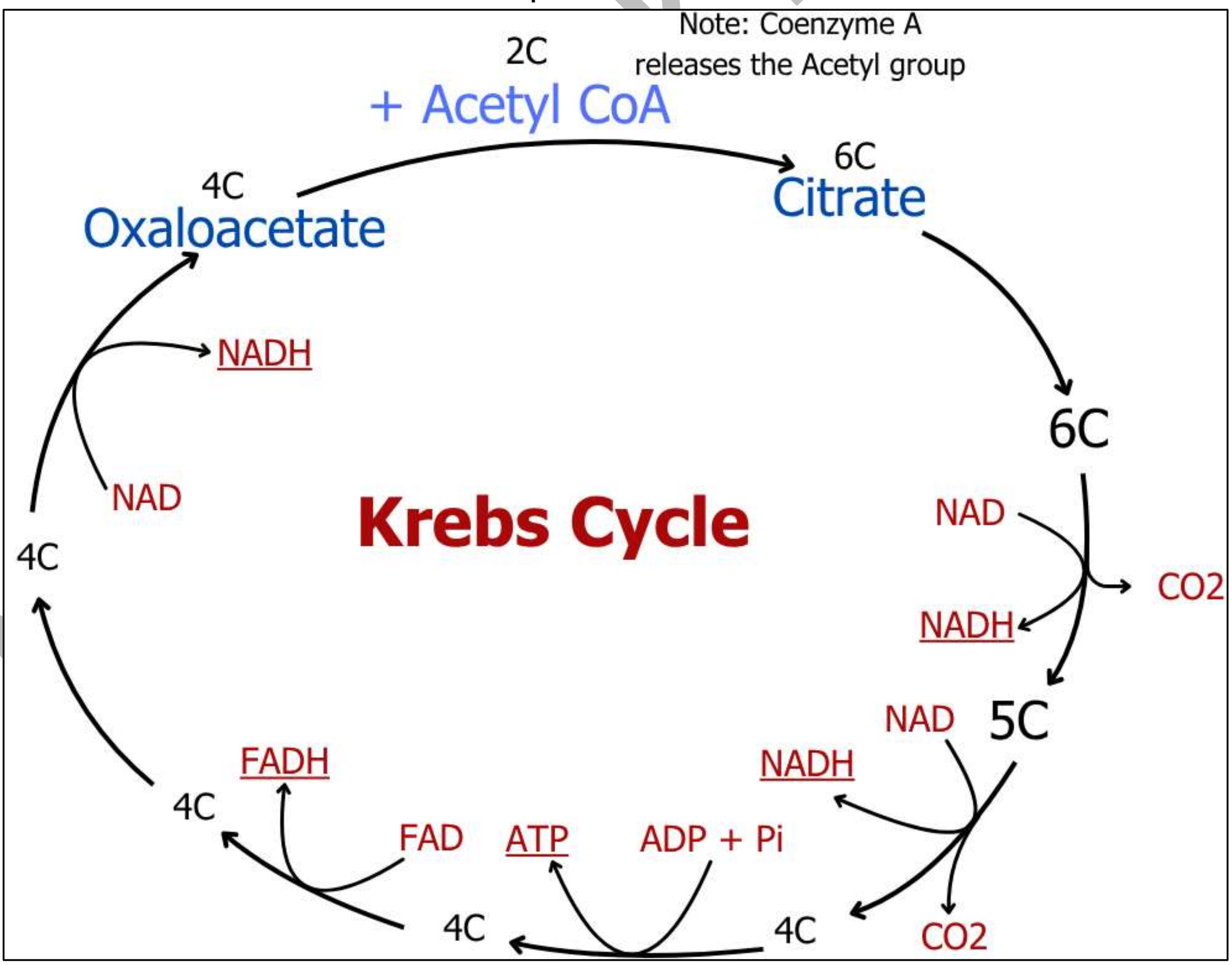
3. **Citrate Formation**: Each Acetyl-CoA (2C) combines with oxaloacetate (4C) to form citrate (6C).
4. **Decarboxylation and Dehydrogenation**: Citrate undergoes multiple steps, releasing 2 $CO_2$ and reducing NAD/FAD to NADH/FADH.
5. **Substrate-Level Phosphorylation**: One ATP is produced per cycle.
6. **Regeneration of Oxaloacetate**: Oxaloacetate is **regenerated** to accept another acetyl group.

**Exam Tip:** Krebs cycle of 2 Acetyl CoA **produces 6 NADH, 2 $FADH_2$, 4 CO2 and 2 ATP**. Oxaloacetate acts as both starting and ending point.

Q: Explain why Krebs Cycle is called a 'cycle'

A: - It leads to **regeneration of Oxaloacetate**, its the starting and the ending point

- All intermediate molecules are present at all times



**Total So Far: 1 Glucose = 10 NADH + 2 FADH + 2 ATP + Oxaloacetate (regenerated) + 4 CO2 (waste gas)**

**Fill This Up!**

| Stage | ATP | $CO_2$ | NADH | $FADH_2$ |
|---|---|---|---|---|
| Glycolysis | | | | |
| Link Reaction | | | | |
| Krebs Cycle | | | | |
| TOTAL | | | | |

# 4. Oxidative Phosphorylation (Inner Mitochondrial Membrane/Cristae of mitochondria)

## 1. Electron Transport Chain (ETC):

- (Reduced)NADH/FADH (from Glycolysis, Link Reaction, Krebs Cycle) Release H atoms at the inner mitochondrial membrane,
  **H splits into Proton and Electron**
- Electrons travel **along** protein complexes (ETC), **releasing energy from excited electrons** to **power proton pump** to pump protons ($H^+$) into the intermembrane space.

## 2. Chemiosmosis:

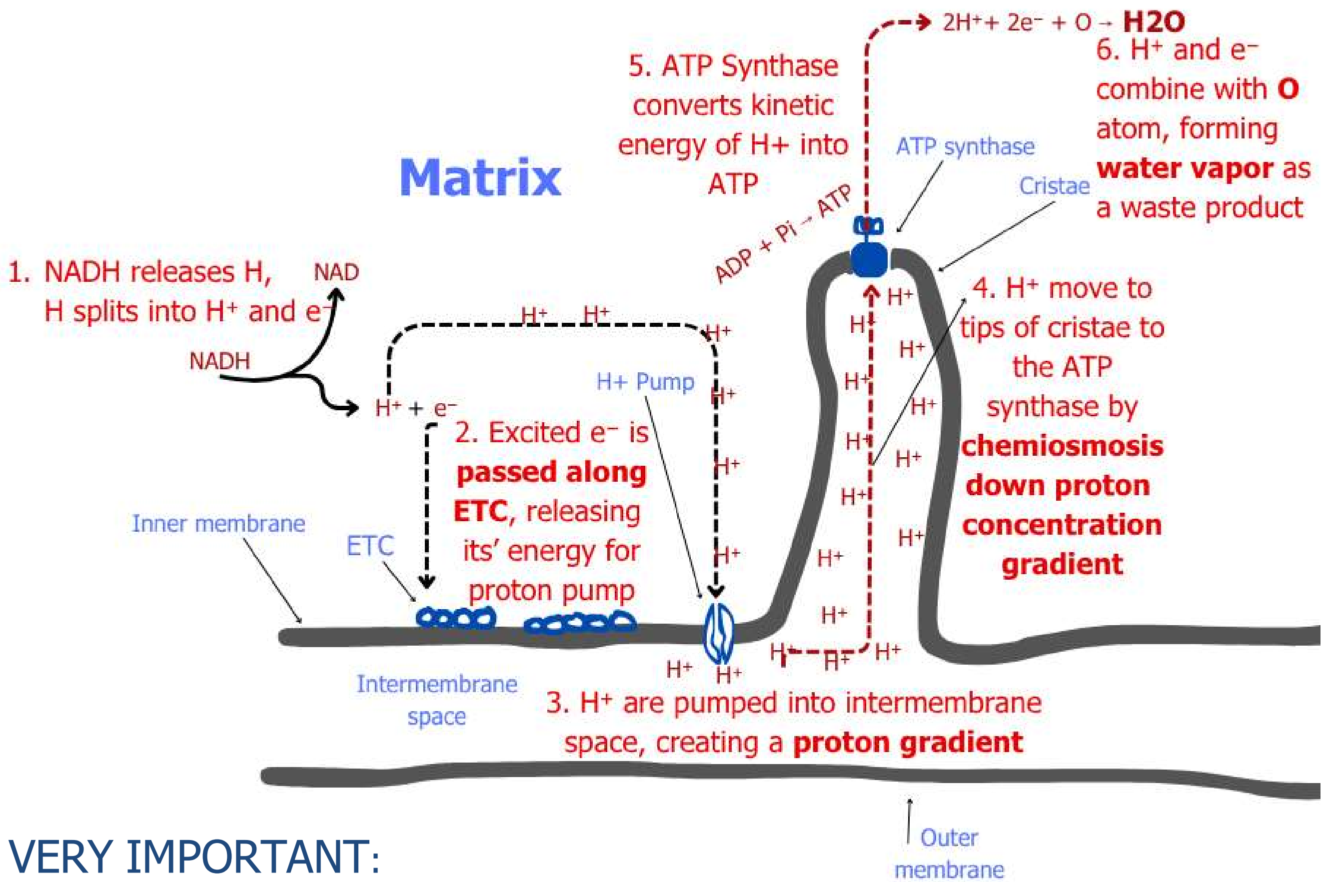
- Protons build up in the intermembrane space, **creating a concentration gradient**.
- Protons flow back through ATP synthase into the matrix, down the concentration gradient by Chemiosmosis (since the **inner/outer membrane are impermeable to protons**), ATP Synthase absorbs kinetic energy of the protons, using it to phosphorylate ADP into ATP (ADP + Pi → ATP)

## 3. Oxygen's Role:

- Oxygen **acts as final electron acceptor** to form water. Without it, the ETC stops, stopping proton pumping and ATP production.

Check Diagram below

# Oxidative Phosphorylation




## VERY IMPORTANT:

Q: Explain how a lack of oxygen (anaerobic conditions) affects the oxidative phosphorylation

**MS Style answer:**

- Since Oxygen is the final acceptor of electrons
- The ETC stops releasing energy from excited electrons **for the proton pump**
- No protons pumped into the intermembrane space, **no proton gradient created**
- No protons move through ATP Synthase, No **Chemiosmosis**, **No ATP Produced**



Q: Describe the main conditions that are required for pyruvate to enter the mitochondrion by active transport.

**Ms Style answer**:

- Mitochondrial membrane to **be impermeable** to pyruvate
- Presence of suitable transport protein
- Oxygen to be available
- ATP for **active transport** against concentration gradient of pyruvate

**Exam Tip:** Oxidative phosphorylation **produces ~28 ATP per glucose**—the main ATP source in aerobic respiration, and Substrate linked phosphorylation only produces 4 ATP:

**1 NADH ≈ 2.5 ATP**
**1 FADH ≈ 1.5 ATP**

1. ***Glycolysis:*** **2 ATP + (2 NADH x 2.5 = 5 ATP) +**
2. ***Link Reaction:*** **2 NADH x 2.5 = 5 ATP) +**
3. **Krebs Cycle: 2 ATP + (6 NADH x 2.5 = 15 ATP) + (2 FADH x 1.5 = 3 ATP)**

**Aerobic respiration total ATP = 2 + 5 + 5 + 2 + 15 + 3 = 32 ATP**

**Q: Explain why the actual net number of ATP molecules produced from aerobic respiration is less than the theoretical number**

- ATP is used to transport pyruvate into the mitochondria by active transport
- Some protons **leak out** of the intermembrane space during oxidative phosphorylation Some kinetic energy of protons is **lost as heat**
- Reduced NAD may be used for other reactions in the mitochondria

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Test Yourself:

**Q: Fill this table up:**

| Stages of Respiration | ATP used | ATP made | Net gain in ATP |
|---|---|---|---|
| Glycolysis | | | |
| Link reaction | | | |
| Krebs cycle | | | |
| Oxidative phosphorylation | | | |
| Total | | | |

ALL IMPORTANT ORGANIC MOLECULES IN AEROBIC RESPIRATION

- Glucose is 6C (Glycolysis)
- Triose phosphate and Glycerate phosphate are 3C (Glycolysis)
- Pyruvate is 3C (produced in Glycolysis, goes into Link Reaction)
- Citrate is 6C (Oxaloacetate + Acetyl CoA $\rightarrow$ Citrate, in Krebs Cycle)
- Acetate/Acetyl CoA is 2C (Link Reaction, goes into Krebs Cycle)
- Oxaloacetate is 4C (gets regenerated in Krebs Cycle)

**Q: Outline the roles of named coenzymes in aerobic respiration.**

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**Q: Explain why the link reaction only occurs when oxygen is available (explain what happens in link reaction and where oxygen is needed)**

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**Q: Describe the processes that occur in the mitochondrial matrix (with ALL keywords mentioned earlier)**

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Now, lets answer the question, Explain Why 1g Fat produce more ATP than 1g Carbohydrates as a respiratory substrate?

- Fats have **more C-H bonds** than carbohydrates, so **more H atoms** than carbohydrates So **more reduced NAD** (NADH) produced
- So **more H atoms released at ETC**, more protons to build up **steeper proton gradient in intermembrane space**
- More **chemiosmosis, more protons passing through ATP synthase**

## Respiratory Quotient (RQ Value)

**Respiratory Quotient: Ratio of volume of carbon dioxide given out divided by volume of Oxygen taken in Per unit time**

Common RQ values:

- Carbohydrates ≈ 1
- Lipids ≈ 0.7
- Protein ≈ 0.9

$$RQ = \frac{\text{Volume of } CO_2 \text{ out}}{\text{Volume of } O_2 \text{ in}}$$

Test Yourself:

This Fig. Summarises the respiration of an organic substance, A, in aerobic conditions. Complete in the spaces provided.

$C_{18}H_{32}O_2$ + ..........$O_2$ ⟶ .......... $CO_2$ + $16H_2O$
substance **A**

**Calculate the RQ for substance A.**

Write your answer to two decimal places. RQ =...............[1]

**Use your calculated RQ value to suggest the group of compounds to which substance A belongs.**

**........................................................................................................ [1]**

***Q: Explain why carbohydrates, lipids and proteins have different relative energy values as substrates in respiration In aerobic conditions. (N18/V2/Q9a)**

- Different substrates have different numbers of hydrogen atoms /C-H bonds Lipids have more C-H bonds in fatty acids than carbs or proteins
- Breaking down of substrates gives H atoms, which reduce NAD/FAD
- Reduced NAD/FAD Release Hydrogen to ETC, which splits into protons and electrons
- More electrons passed to ETC = more energy released to pump protons
- More protons = steeper proton gradient = more chemiosmosis = more ATP produced from lipids

# Anaerobic Respiration

Occurs in Anaerobic conditions (absence of oxygen)

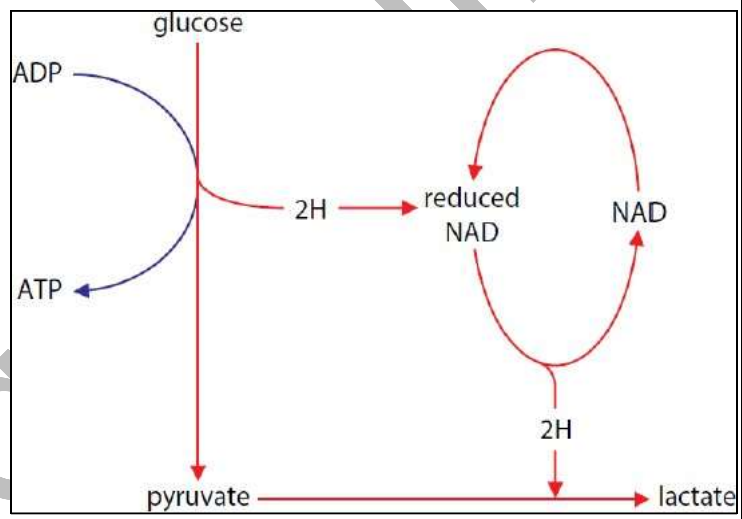
- No **Final Acceptor of electrons** and protons
- NAD Can't be regenerated from reduced NAD
- Pyruvate **doesn't** enter the mitochondrion, No link reaction, no Krebs cycle, no oxidative phosphorylation

So anaerobic respiration is completely based on **Glycolysis** **i**n the cytoplasm only

*Anaerobic respiration has 2 pathways

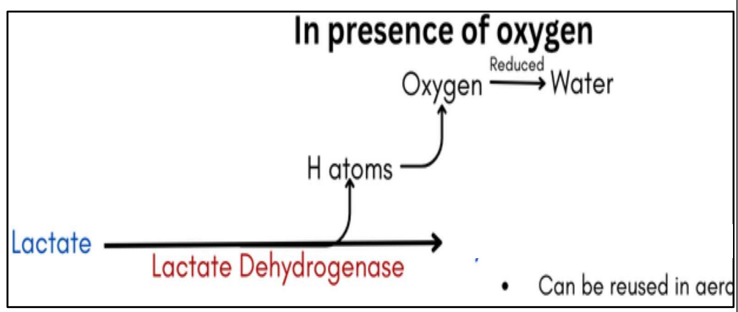
Animal/Bacteria Cells (**Lactate Pathway**):

1. Pyruvate accepts electrons from NADH to form lactate. Building up oxygen debt
2. Regenerates NAD To be able to continue glycolysis again
3. But creates oxygen debt to convert the lactate back into glucose by reoxidising it
4. Produces 2 ATP from glycolysis (very little amount of energy from 1 glucose)



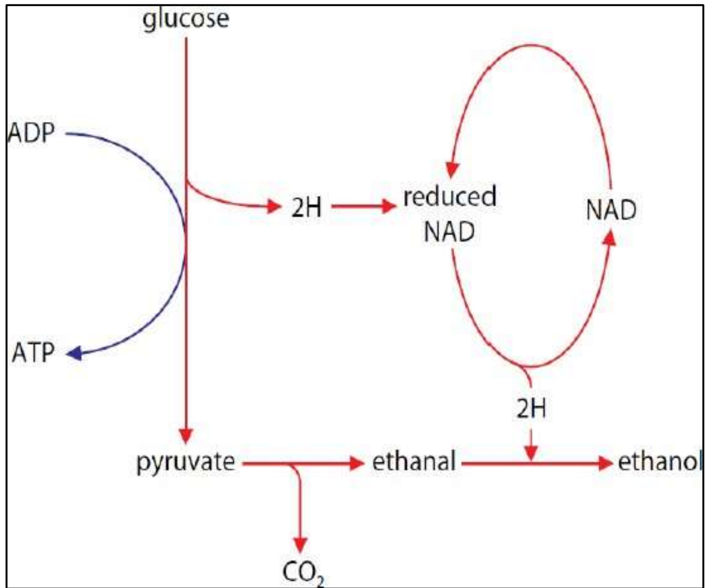

When Oxygen is present:

- **Lactate Dehydrogenase** oxidises Lactate back into pyruvate
- **Oxygen accepts the H atoms from the lactate and gets reduced into H2O**
- So, Lactate pathway anaerobic respiration is reversible



Yeast/Plant Cells
(**Ethanol Pathway / ethanol fermentation**):

1. Glucose goes through Glycolysis, converted to Pyruvate, producing **reduced NAD and 2 ATP**
2. Pyruvate gets **decarboxylated** converts to **ethanal**,
3. Reduced NAD releases H and **reduces ethanal to ethanol**
4. Releases $CO_2$ and **regenerates NAD to continue glycolysis**.




Ethanol **cannot be converted back to Pyruvate**, No oxygen debt

## Oxygen debt:

Q: Why is more oxygen required in the body after a session of vigorous excercise?

- To reoxidise Lactate in the liver back to pyruvate
- To reconvert pyruvate back into glucose
- To **reoxygenate haemoglobin**

### Differences between Lactate pathway and Ethanol Pathway

| Lactate Pathway | Ethanol Pathway |
|---|---|
| Takes place in Animal and Bacteria cells | Takes place in Yeast and Plant cells |
| Pyruvate Reduced to Lactate (one step process) | Pyruvate Reduced to Ethanal, THEN Ethanal is reduced to Ethanol (Two step process) |
| No Decarboxylation | Decarboxylation of Pyruvate first |
| Reversible Reaction | Irreversible |
| Lactate Dehydrogenase **Can oxidise lactate into Pyruvate** | Ethanol Dehydrogenase **Can oxidise Ethanol into Ethanal only** |

**Exam Tip: Anaerobic respiration primarily regenerates NAD. ATP comes only from glycolysis by substrate linked phosphorylation.**

## Test yourself:

**Q: Explain why less ATP is produced when yeast respire in anaerobic conditions compared to when yeast respired in aerobic conditions**

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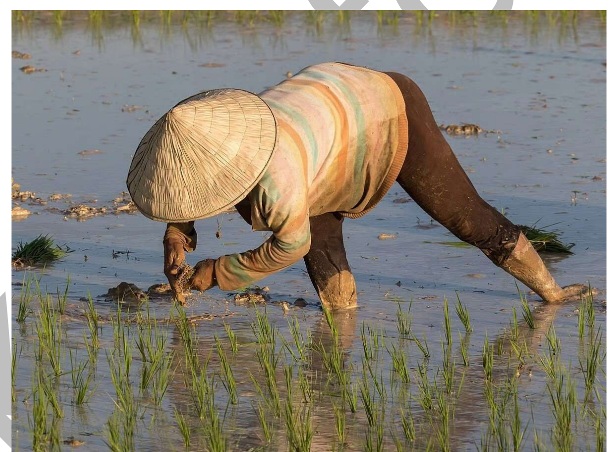
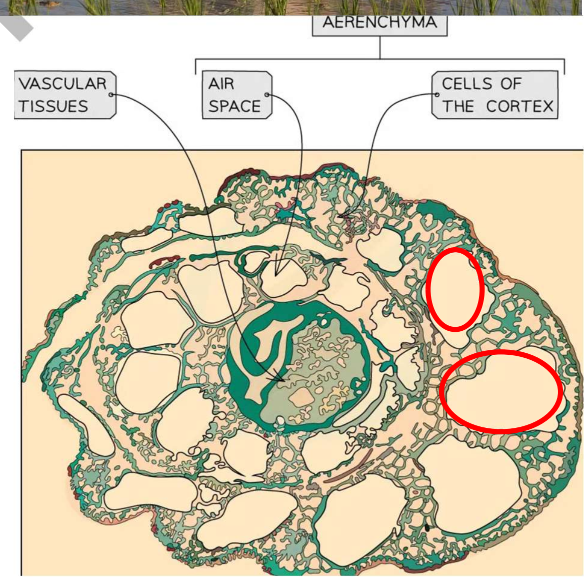
## Adaptations of Rice plants to survive in flooded areas



**As the water rises in plants fields, plant roots don't get enough oxygen that they need for aerobic respiration, plant leaves don't get the $CO_2$ they need for Photosynthesis.**

**Rice plants have adapted to surviving in these conditions, as they have:**

- **Increased upwards growth rate** so that the leaves remain above water and can access Oxygen and $CO_2$
- **Aerenchyma tissue** in stems and roots
  - **A** tissue of Air spaces **spreading throughout the plant**
  - **Allows** gases entering the Stomata to travel to the stem and roots **that are submerged underwater, allowing continuing of** aerobic respiration
- When **Aerobic isn't enough**, rice plant tissues can undergo ethanol fermentation
  - As they tolerate the high levels of toxic ethanol
  - Due to High concentrations of Ethanol Dehydrogenase, which breaks down ethanol




## End Of Chapter